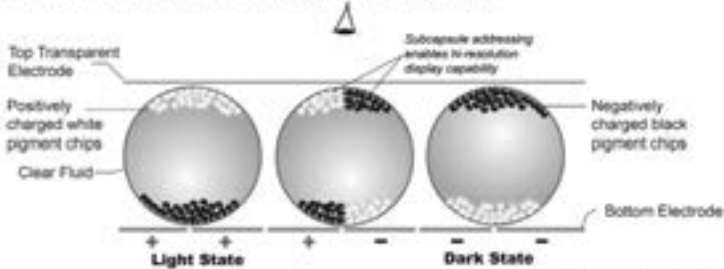


*Cross-Section of Electronic-Ink Microcapsules*

NOTE: Copyright E-Ink Corporation, 2002. Images not drawn to scale - for illustration purposes only.

E - I N K

is even longer. Needless to say, the market is waiting impatiently to see some progress in this field, and researchers are working overtime to meet this demand.

E-Ink's technology is similar to what Sheridan came up with at Xerox PARC. Instead of using microcapsules that are coloured half-black and half-white, E-Ink uses microcapsules that contain granules of a positively-charged white pigment and granules of a negatively-charged black pigment.

These capsules are suspended in a clear fluid that lies between thin layers of a plastic material, which is, in turn, laminated on a layer of micro-circuitry and electrodes.

Now, in order to draw a display, the electrodes are charged either positive or negative, according to the image requirement. The micro-circuitry is small enough for two electrodes to be placed under each microcapsule—if both

electrodes are charged positively, the positively-charged white pigment granules in the capsule directly above them are pushed to the top, thus forming a white dot, or pixel. Similarly, a negative charge causes the negatively-charged black pigment granules to be pushed up, thus forming a black pixel.

Now, the two electrodes under each microcapsule can be polarised differently as well, thus forming a half-white and half-black pixel. This is how E-Ink's technology achieves higher resolutions and apparent shades of grey. All this circuitry forms a screen with a calculable resolution, which can then be controlled by a display driver or graphics chip.

Moreover, the "E-Ink" is reflective in nature, and it looks just like print on paper, has a readable angle of 170 degrees, and is clear even in direct sunlight.